Jasper Sands

Palo Alto, CA | (650) 924-8429 | jaspersands02@gmail.com | jaspersands.com | github.com/Jaspersands | linkedin.com/in/jaspersands

Quantum algorithms, error correction, and machine learning for qubit control. MS in Computer Science at Columbia. Thesis at Diraq and UNSW Sydney on automating silicon spin qubit measurement.

EDUCATION

Columbia University, MS in Computer Science, Quantum Computing Track · GPA 3.95/4.0 — Sep 2025 – May 2027
Quantum coursework: Error Correction · Engineering · Simulation & Computing Lab · Optimization & ML.
TA: Introduction to Quantum Computing.

Washington University in St. Louis, BS in Computer Science and Economics · GPA 3.81/4.0 — Aug 2021 – May 2025
Coursework: Machine Learning · Analysis of Algorithms · Computer Security. **TA:** Malware Analysis; Parallel Programming.

SELECTED PROJECTS

WhatTheDuck: quantum amplitude estimation for financial Value at Risk · *CUDA-Q, C++* — Overall Winner, MIT iQuHACK 2026
- Wrote the GPU-accelerated state prep, threshold oracle, and bisection search driving Iterative Quantum Amplitude Estimation.
- Measured query scaling matched the $O(1/\varepsilon)$ bound against an $O(1/\varepsilon^2)$ quasi-Monte Carlo classical baseline.

WASM QEC Simulator: surface-code decoding in the browser · *Rust, WebAssembly* — qcompiler.jaspersands.com
- Symplectic stabilizer simulator for rotated and XZZX surface codes, with exact MWPM and Union-Find decoding client-side.
- Finite-size scaling puts the threshold at 3.2% with MWPM, 2.7% with Union-Find, and 14.7% under code-capacity noise.

Q-Search: proof-gated quantum algorithm research and verification engine — qsearch.jaspersands.com
- Automated search for structural quantum advantage on non-abelian HSP, dihedral coset, and linear code equivalence problems.
- Formal proof gates test each mechanism against 1,200+ classical baselines across 720+ theorem modules, indexing negative results.

LLM-Based Lossless Text Compression · *PyTorch, CUDA* — jaspersands.com/llmcompression
- Arithmetic coding on LLM predictions, 5–16× average and 39× best case, at 100–1000× lower throughput than gzip and zstd.

RESEARCH

Diraq / UNSW Sydney · *MS Thesis Research* — Sep 2026 – May 2027
- Automating silicon spin qubit characterization under Dr. Nard Dumoulin Stuyck, replacing manual tuning and interpretation.
- Acquiring charge measurements from cryogenic probing, and classifying device states against what an expert would call by hand.

Cidon Systems Research Lab, Columbia University · *Graduate Researcher* — Sep 2025 – Jun 2026
- Built phishing and spam detection pipelines with Dr. Asaf Cidon and Barracuda Networks, combining language-model embeddings with production email telemetry and tuning for precision against a fixed false-positive budget.

Complex Resilient Intelligent Systems Lab, Columbia University · *Graduate Researcher* — May – Dec 2025
- Reduced hallucination in generative protein modeling under Dr. Venkat Venkatasubramanian, grounded in molecular graph data.

Foster Care Policy Database, Washington University in St. Louis · *Undergraduate Researcher* — Sep 2024 – May 2025
- Fine-tuned LLaMA-3 with Hugging Face for QA over 50k foster-care policy documents, with human-in-the-loop feedback.

EXPERIENCE

Smack Technologies · *AI Intern* — May – Aug 2026
- Built physics and sensor simulation, integrating the knowledge graph to expand corpus generation.
- Benchmarked open-weight models head to head on speed, intelligence and concurrency, speeding corpus generation 64x.
- Bounded simulator error to 0.55% against references outside its own tests, in a report that regenerates when the physics changes.

Highnote · *Cybersecurity Engineer Intern* — May – Jul 2024
- Designed and deployed the company-wide SIEM for a card-issuing fintech, unifying AWS, GCP, and Datadog telemetry into one Elasticsearch cluster ingesting over 1 TB/day, and wrote 100+ rule-based and ML anomaly detections.

Mindtrip · *Software Engineer Intern* — May – Jul 2023
- Built internal admin and secure-query tooling in JavaScript and Rails via Retool, saving roughly 250 engineering hours a month.

HONORS & LEADERSHIP

- **Overall Winner** and **NVIDIA Ecosystem Award**, MIT iQuHACK 2026 · **Audience Favorite**, Harmoniqs Quantum Design 2026 · **Runner-Up**, Qualcomm Snapdragon Multiverse 2026 · **Qualified**, NYU Abu Dhabi Hackathon for Social Good 2027
- **Graduate Lead**, Columbia Quantum Algorithms Reading Group: weekly sessions through all 33 chapters of Andrew Childs' notes.

SKILLS

Quantum tools: Qiskit and Qiskit Runtime, Cirq, PennyLane, CUDA-Q, Stim, PyMatching, QuTiP
Quantum methods: Hamiltonian simulation, phase and amplitude estimation, QSVT and block encoding, LCU, VQE, QAOA · randomized benchmarking, tomography, noise modeling · quantum complexity (BQP, QMA), query complexity
Languages: Python, C++, Rust, C, MATLAB, JavaScript, TypeScript, Go, Java
ML & systems: PyTorch, CUDA, Hugging Face · Linux, Git, Docker, WebAssembly, AWS, GCP, Elasticsearch, LaTeX